INCH-POUND

A-A-54867
14 May 1993

COMMERCIAL ITEM DESCRIPTION

OTOSCOPE AND OPHTHALMOSCOPE SET

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-O-36811A, which is cancelled.

This Commercial Item Description covers a miniature otoscope with fiber optics illumination and a miniature ophthalmoscope both contained in a single pocket-size carrying case suitable for examination of the ears and eyes.

Salient characteristics:

General. Shall be a miniature otoscope and ophthalmoscope set, diagnostic, capable of fitting in a pocket and suitable for examination of the eyes and ears.

Material. All glass parts of the set shall be fabricated from high quality optical glass. All other materials shall be as specified herein and of high quality throughout, suitable in all respects for the purpose intended.

Shall be a pocket-size ophthalmoscope and fiber optics illuminated otoscope set consisting of the following components:

Component	Quantity
Fiber optics otoscope head with lamp	1
Battery handle for otoscope head	1
Ophthalmoscope head with lamp	1
Battery handle for ophthalmoscope head	1
Spare lamp for otoscope	1
Spare lamp for ophthalmoscope	1
Reusable specula: 2, 3, 4 and 5 millimeters(mm)	1 each
Adult-size disposable specula	10
Cushioned and compartmented carrying case	1

AMSC N/A FSC 6515

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Otoscope head. The otoscope head shall be of the fiber optics type. The otoscope head shall be fabricated from nylon or other suitable high-impact plastic. The head shall have a permanently attached metal speculum slotted to accept and hold the specula. The glass fiber bundle shall be encased between the funnel-shaped speculum holder and the funnel-shaped inner part. The speculum holder shall be fabricated of metal, with the surfaces highly polished and chromium plated or fabricated of stainless steel. The interior funnel shall be fabricated of suitable black plastic with matte or dull surface on the interior, to prevent undesired light reflections. The distal end of the body (metal speculum holder) shall extend beyond the distal end of the fiber optics light transmitting bundle for a distance of not less than 2.5 mm. As an alternate, the distal end of the body shall be flush with the fiber optic light transmitting bundle. The source of illumination shall be a lamp as described herein. The axial illumination provided by the otoscope head with an annular fiber optics light transmitting bundle shall be not less than 30 lumens per square foot at the distal end of the longest speculum at a lamp voltage of not more than 2.5 volts. The head shall have a threaded connector which attaches to the battery holder handle. The head shall be equipped with a magnifier that shall be capable of being moved out of the line of vision to permit oblique observation. The magnifier shall have a focal length of not more than 85 mm for obtaining sufficient magnification. The inner diameter of the viewing aperture shall be not less than 4 mm.

Lamp. The otoscope lamp shall be rated at a maximum of 280 milliamperes (ma) at 2.2 volts. As an alternate, the lamp shall be a halogen type rated at 660 ma at 2.5 volts. The lamp shall be a plug-in type with center contact. The base of the lamp shall contain a spring element to exert a pressure on the wall of the lamp socket to prevent the lamp from falling from the socket. A lamp shall be furnished in position in the otoscope head.

Specula.

Reusable specula. The reusable aural specula shall be supplied in four sizes: 2, 3, 4 and 5 mm inside diameter (ID) at the distal end. These specula shall be fabricated of a suitable plastic material, black in color, and nonreflective to prevent interfering glare to the operator. The specula shall be designed to permit ready attachment to, or removal from, the otoscope head.

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Disposable specula. Each set shall be equipped with ten disposable specula of the adult size fabricated from a suitable plastic. The specula shall be contained in the case in a transparent plastic container.

Ophthalmoscope.

Ophthalmoscope head. The ophthalmoscope shall be of the mirror type with single rekoss disc (lens disk). The head or housing shall be fabricated from a suitable plastic material. The exposed surfaces shall be black. The viewing lens shall be enclosed on the viewer's side with an optically-polished face plate. As an alternate, the viewers side shall be open. The light emitting aperture shall be capable of being closed with a sliding cover. The ophthalmoscope head shall be designed for ready attachment to, or removal from, the battery handle. The connector for attaching the ophthalmoscope head to the battery handle shall be fabricated of a suitable metal. The optical reflecting system, the device for selecting the filter and apertures, and the rekoss disc shall be incorporated into the head.

Optical reflecting system. The optical reflecting system shall consist of a lamp, a condenser lens, a device for selecting the aperture, an illumination objective for projecting the apertures at a suitable distance from the ophthalmoscope, and a mirror for deflecting the beam of light to the lens system. The aperture shall include a full circle, a half circle, a vertical slit, and a green filter. As an alternate, the aperatures shall include a full circle, a small full circle, a slit, a green (red free) filter and a fixation pattern. The device for selecting the apertures and filters shall be designed to permit convenient operation without interrupting the examination. The lamp shall be located in the connector of the ophthalmoscope head. Indexing of the lamp receptacle shall be provided so that the lamp filament is always properly centered and focused in relation to the optical system. No adjustment shall be required when replacing a lamp. No special tool shall be required to replace a lamp. A spring element shall be provided in the lamp housing to provide a frictional force on the lamp to prevent the lamp from sliding from its receptacle. The illumination provided by the optical system, at a point 7-3/4 inches from the exit port of the ophthalmoscope, shall be not less than 3.5 lumens per square foot when the lamp is operated at 2.5 volts. The illuminated spot produced by the ophthalmoscope head shall be uniform.

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Rekoss disc. The rekoss disc shall consist of 16 lenses, 0 to +20 diopters minimum and 0 to -20 diopters minimum with an adequate working assortment of powers. Numeric markings shall be provided in the rekoss disc to identify the lens selected. The numerals shall be clearly visible in a magnified and illuminated window in the ophthalmoscope head. Magnification and illumination shall be sufficient to enable the operator to read the markings in a darkened room. Plus (+) diopters shall be marked in black, minus (-) diopters shall be marked in red. The lenses shall be fabricated of suitable polished, optical glass or optical grade methyl methacrylate. The lenses shall be securely mounted to the rekoss disc. A click stop shall be provided for holding each lens in position.

Lamp, ophthalmic. The lamp shall be the plug-in type with a center contact. The base of the lamp shall be equipped with an indexing device for positioning the lamp in relation to the optical system. The lamp shall have a closed coil filament. The axis of the filament shall be straight. The lamp shall draw a maximum current of 280 ma when operated at 2.2 volts. As an alternate, a halogen lamp shall be provided which shall draw 660 ma when operated at 2.5 volts.

Battery handles. The battery handles shall be cylindrical in shape and of a size to hold two AA or C batteries. The battery handle shall be fabricated from nylon or other suitable high impact plastic. The handle shall have a maximum length of 12 $\pm$ .12 centimeters (cm), excluding the heads, and shall be 20 $\pm$ 2 mm outside diameter (OD) at the top and 18 $\pm$ 2 mm OD at the bottom. The upper end of the handle shall have a threaded connector for easy attachment of the otoscope and ophthalmoscope heads. The other end of the battery handles shall be provided with a threaded cap. The cap shall be provided with a contact spring designed to hold the batteries securely in position while making electrical contact with the cap. Each battery handle shall be equipped with a metal pocket clip suitable for securely holding the handles in the pocket with the otoscope and ophthalmoscope attached. The pocket clip shall include, as an integral part, a sliding switch, which switches off the handle when clipped into a pocket. As an alternate, the switch shall be a rotating type actuated by the thumb and forefinger. The battery handles shall be supplied without batteries.

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Carrying case. The carrying case shall be fabricated of a high impact material. There shall be a form-fitting insert compartment to accommodate all components of the set. The case shall be provided with a continuous hinge with suitable springs which shall automatically close the cover under spring pressure. The top and bottom sections of the case shall have metal-trimmed edges. The overall dimensions of the carrying case shall be 6-5/8 ± 1/4 inches long, 3-1/4 ± 1/4 inches wide by 1-1/8 ±1/4 inches thick.

Identification marking. Each instrument, the ophthalmoscope head, the otoscope head, the battery handles, and the carrying case shall be permanently and legibly marked with the name or registered trademark of the manufacturer.

Instructions. Each set shall be furnished with two copies of clear instructions for assembly and use of the instrument. The instructions shall contain maintenance information and a list of component parts including the manufacturer's part or catalog number.

Workmanship. The otoscope and ophthalmoscope set shall be free from defects which detract from its appearance or impair its serviceability. Components shall be free from flash, rough areas, cuts, distortion, dust or foreign matter.

Unit. Set (SE). One otoscope and ophthalmoscope set, as specified, constitutes one unit.

QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspection are deemed necessary to assure supplies and services conform to prescribed requirements.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Examination. Examination shall be conducted to determine compliance with specification requirements.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as specified. Unit of product for examination purposes shall be one otoscope and ophthalmoscope set as specified. Sampling shall be inspection level II with an Acceptable Quality Level (AQL) of 1.0.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for determination of two or more test characteristics.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and as specified. Unit of product for test purposes shall be one otoscope and ophthalmoscope. Sampling shall be inspection level S-2 with an AQL of 1.0.

Lamp life. The lamp shall be supplied with 2.2 ± 0.125 volts direct current. The lamp shall be operated in cycles of 5 minutes on and 2 minutes off. Inspection level shall be S-1 with an AQL of 1.5. The lamp shall have a minimum life of 12 hours.

Boil test. The reuseable specula shall be boiled in distilled water for four successive 15-minute periods. After each 15-minute period, the specula shall be removed and allowed to air dry for 10 minutes. There shall be no discoloration or dulling of the surface or other defects. Inspection level shall be S-1 with an AQL of 2.5.

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Alcohol immersion test. The reuseable specula shall be completely immersed in 95 percent ethyl alcohol for 30 minutes. After removal, the specula shall be rinsed thoroughly in distilled water and dried. There shall be no discoloration or dulling of the surface or other defects. Inspection level shall be S-1 with an AQL of 2.5.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

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Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order.

NOTE: The following National Stock Number and its respective Item Identification is covered by this document:

NATIONAL STOCK NUMBER	ITEM IDENTIFICATION
6515-00-165-6544	OTOSCOPE AND OPHTHALMOSCOPE SET, Battery Type, Pocket Size

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project Number: 6515-5549
Location: c:\word\aoc\nisn\otoscop2

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1 The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given.

2 The submitter of this form must complete blocks 4, 5, 6, and 7

3 The preparing activity must provide a reply within 30 days from receipt of the form

NOTE This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER A-A-54867	2 DOCUMENT DATE (YYMMDD) 14 May 1993

3 DOCUMENT TITLE
CTOSCOPE AND OPTHALMOSCOPE SET

4 NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed.)

5 REASON FOR RECOMMENDATION

6 SUBMITTER

a NAME (Last, First, Middle Initial)	b ORGANIZATION
c ADDRESS (Include Zip Code)	d TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7 DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a NAME	b TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Defense Personnel Support Center ATTN. DPSC-MSE	(215)7372870	444-2870
c ADDRESS (Include Zip Code) 2800 South 20th Street Philadelphia, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike Suite 1403 Falls Church VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289 2340	

DD Form 1426, OCT 89 Previous editions are obsolete 198-29

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